

These hybrid off-shore solar cells are said to require minimal maintenance and generate higher energy yields than ever before.

Waterproof phones

Your phone drops in water, no worries! A technology developed by Utah-based company HzO will protect your device.

Dubbed as WaterBlock, the technology works by coating the electronics at the molecular level. Thus it protects the device from the inside, out. The eco-friendly nanotechnology-based vapour coating is composed of a non-toxic organic material. The chemical vapour deposition blankets vital electronic circuitry with a nano-thin film containing highly effective, water-repelling properties.

WaterBlock – when applied to clean, moisture-free surfaces, including plastic, metal, ceramic, steel or tin – is transparent and creates a finish that doesn't disrupt the look and feel on whatever it is applied. The technology is designed to shield electronics not only from water or any liquid but from humidity as well.

Magnetic memory bit using just twelve atoms

IBM researchers have developed a magnetic memory bit that can store information in as few as twelve magnetic atoms. The magnetic storage is at least a hundred times denser than today's hard-disk drives and solid-state memory chips.

Using a scanning tunneling microscope (STM), scientists atomically engineered a grouping of twelve anti-ferromagnetically coupled atoms that stored a bit of data for hours at low temperatures. Taking advantage of their inherent alternating magnetic spin directions, they demonstrated the ability to pack adjacent magnetic bits much closer together than was previously possible. This greatly increased the magnetic storage density without disrupting the state of neighbouring bits.

High-performance thin-film boost for electronics research

A research group led by Dr Minoru Osada and principal investigator Dr Takayoshi Sasaki of the International Center for Materials Nanoarchitectonics (MANA) at the National Institute for Material Science (NIMS) in Japan claims to have developed the world's highest-performance thin-film capacitors using a high-permittivity (high-k) dielectric sheet with molecular-level thickness (~1 nm).

The thin films are based on titanium-niobate nanosheets as building blocks. Scientists delaminated layered oxides and stacked sheets on an atomically flat SrRuO₃ substrate creating films between 5 and 15nm thick with permittivity of 160-300. In these nanosheets, the octahedral distortion inherent to site engineering by niobium (Nb) doping results in a giant molecular polarisability.

New cooperative functions that originate from the mutual interactions between nanoscale structural units comprise the focus of nanoarchitectonics. This research demonstrates simultaneous improvements in a number of material properties, including relative permittivity, lower loss and leakage current.

Retrofit PowerLED Tube Light



The BINAY PowerLED Retrofit tubelight fits directly into a standard fluorescent tube luminaire or strip

- Wattage less than 50% of fluorescent system, with similar light output
- Operates from 100VAC to 260VAC (No flickering at lower voltages)
- Provides sharper visual clarity than fluorescent tube
- Life of 12 to 15 years, with a warranty of 5 years



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INVENTING NEW TECHNIQUES OF PRODUCING LIGHT

BINAY PowerLED Dome Type High Bay Light



**Saves Money
Saves Energy
Better, Clearer, Spectrally Enhanced Light
Designed for Long and Reliable System Lifetime**

BINAY's PowerLED Dome Type High Bay Luminaires are a one-to-one replacement for existing HID High Bay Luminaires, and offer low power consumption and a maintenance-free long life in an economical form factor

- Saves 65-70% power consumption over Metal Halide or HPSV
- Life of 10 to 15 years eliminates regular bulb and ballast replacement costs
- Instant turn-on to full brightness
- Full spectrum white light provides greater clarity and visual acuity; improves visibility and safety



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